



DPUPolynom

Computes the value of a polynomial function of the input value *In*.

1. Input:

In Input value

2. Dynamic parameters:

<i>A[0..5]</i>	Coefficients of the polynomial (default: 0)
<i>Multiplicator</i>	Additional gain of the output value (default: 1)
<i>Divisor</i>	Additional divisor of the output value (default: 1)
<i>Offset</i>	Additional offset of the output value (default: 0)
<i>InMin</i>	Minimal permitted value for the input value (default: no limitation)
<i>InMax</i>	Maximum permitted value for the input value (default: no limitation)
<i>OutMin</i>	Minimal permitted value for the output value (default: no limitation)
<i>OutMax</i>	Maximum permitted value for the output value (default: no limitation)

3. Outputs:

Out Output value

4. Functional principle:

In the first step, **DPUPolynom** limits the input value in the following way:

$$x \leftarrow \begin{cases} InMin & | & In < InMin \\ In & | & InMin \leq In \leq InMax \\ InMax & | & In > InMax \end{cases}$$

After this, the polynomial function is computed:

$$y \leftarrow A0 + A1 \cdot x + A2 \cdot x^2 + A3 \cdot x^3 + A4 \cdot x^4 + A5 \cdot x^5$$

Next, the parameters *Multiplicator*, *Divisor* and *Offset* are taken into account:

$$o \leftarrow \frac{y \cdot Multiplicator}{Divisor} + Offset$$

Finally, the output value is limited:

$$Out \leftarrow \begin{cases} OutMin & | & o < OutMin \\ o & | & OutMin \leq o \leq OutMax \\ OutMax & | & o > OutMax \end{cases}$$



5. Example:

The following script is an example for using DPUPolynomial:

```
DPUPolynomial polynomial
{
    Computer = {LOCALHOST};
    Index = 1;

    In = 2;

    A0 = 1;
    A1 = 2;
    A2 = 3;
    A3 = 4;
    A4 = 5;
    A5 = 6;

    Multiplier = 10;
};
```

The given script computes the following polynomial:

$$y \leftarrow 1 + 2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + 5 \cdot 2^4 + 6 \cdot 2^5$$

Finally, y is multiplied by 10.